Phase 3D | Component integration and lighter chassis

PULSIPHER FARMS / MOAPA VALLEY CORN MAZE

Resolve weight and fit together

PF-TRAIN-S3D-R0 | 18 October 2026 UTC

Preliminary engineering development - not for fabrication, construction, procurement release or passenger operation. This issue develops alternatives and specific acceptance inputs for the first coach. It does not represent manufacturer project approval, a completed occupied mockup or a professional design release.

The retained basis is a diesel locomotive with three regular coaches and one separate accessible coach, 36-inch evaluation gauge, enclosed heating/cooling, hydraulic service-brake preference and the owner's **0% longitudinal grade** target. North Pole location remains deferred; passengers disembark there. The technical site excludes business projections and ride-experience planning.

Decision advanced	What this phase contributes
Chassis mass	Four coordinate-based member alternatives with a consistent material takeoff and explicit structural/load-path comparisons.
Bogie and floor fit	Documented running-gear references, configuration gaps, load distribution and dimensional returns that govern the provisional floor height.
Floor and body	The mass and stiffness consequences of changing support spacing and panel thickness; a ledger that keeps incomplete body assemblies visible.
Brake and power integration	Wheel/brake torque requirements, adhesion and hydraulic response inputs, equipment electrical envelopes and functional control separation.
Review handoff	Coordinated drawings, rotating comparison models, editable calculations, component returns and first-article evidence requirements.

01 | Controlled continuation of Phase 3C

The current saved Phase3C report and public technical site version 5 were checked before starting this revision. The previous files remain available. No newer Phase3D design was found in the available files at the start of this work.

Disposition	Controlled treatment
Retain	306 x104 in clear cabin; 312 x110 in outer body; 400 in frame study length; bogie centres x45/261 in; 28 in finished-floor target. The 32-place regular interior remains pending occupied trials.
Retain	7,040 kg regular and 7,535 kg accessible tare allocations; 54,755 kg nominal and 59,840 kg stress train cases; 13,200/11,000 kg individual-car upper calculation cases and 65,100 kg train upper case. These are not supplier ratings.
Revise the comparison	Do not regard the 1,913.259 kg Phase3C member model as a selected chassis. Compare a lighter section/grid arrangement against both structural consequences and omitted connection/attachment mass.
Coordinate explicitly	A thicker panel changes the support plane when finished floor is fixed. A shallower main girder does not raise the retained 12-inch centre bolster. Body, bogie and boarding dimensions must be revised together.
Reject unsupported continuity	A catalogue axle rating, common track gauge, brochure image or nonintersecting model is insufficient to select equipment. Prior allowances remain targets until reconciled with supported configured masses.
Still open	Actual occupied mockup results; configured supplier drawings and ratings; responsible engineer's criteria and released connections; final survey/alignment and approval pathway.

All four frame models retain the 3/4-inch floor-panel study so their geometry can be compared directly. Any thicker-panel alternative is identified separately in the calculations and drawings. Neither a lower bare-steel mass nor a favorable bending-only floor result is a construction acceptance.

02 | How to use the comparison

Review package	Decision and evidence required
D0 - retained benchmark	Use the Phase3C steel grid as a common reference for mass, support locations and load paths. It has no production release.
D1 - lighter members, same grid	Check thinner-member bending, stiffness, local connections, corrosion allowance, fatigue, distortion and obtainable material before accepting the mass reduction.
D2 - wider grid, deeper mains	Check the longer floor spans and seat-rail spans, concentrated wheel/seat loads and crossmember reactions. Confirm panel support/edge requirements before accepting removed transverse members.
D3 - wider grid, shallower mains	Carry all D2 conditions and the changed primary-girder stress/stiffness. The retained 12-inch bogie bolsters still control the low central support interface.
Engineered purchased undercarriage	Request the exact complete assembly and vehicle-level scope, including gauge/wheels, suspension, frame/body interfaces and brakes. Compare its measured mass, load cases and mounting geometry with the same coach requirements.

The lowest steel takeoff is not automatically the best vehicle. Panel/hardpoint reinforcement, fatigue details, mounting brackets, bogie mass, body framing and inspection access can change the ranking. Require one coordinated configuration with a closed mass ledger, not a collection of individually attractive catalogue parts.

03 | Lighter chassis alternatives and load paths

The useful reduction is a coordinated frame and floor change. **D2 removes 386 kg from the Phase 3C bare-member benchmark while retaining the 12-inch main girders.** It has 24-inch cabin crossmember spacing, so floor qualification and revised joints are prerequisites. D3 saves another 103 kg by using 10-inch main girders, but increases their elastic bending deflection by 59% under the same load. None of these arrangements demonstrates the earlier 1,500 kg finished-frame allowance.

Retain 0% longitudinal grade, 36-inch gauge as an evaluation basis, the 7,040 kg regular-coach tare allocation and the 13,200 kg upper structural study case. No capacity, rated gross weight or material purchase is approved by this comparison.

1. Four mutually exclusive alternatives

ID	Cabin grid / main girders / secondary members	Modeled members	Bare steel	Change from D0
D0	16 in / HSS 12×4×¼ / HSS 4×2×3/16	166	1,913.26 kg	Retained benchmark
D1	16 in / HSS 12×4×¼ / HSS 4×2×⅛	166	1,660.67 kg	−252.58 kg
D2	24 in / HSS 12×4×¼ / HSS 4×2×⅛	110	1,527.29 kg	−385.97 kg
D3	24 in / HSS 10×4×¼ / HSS 4×2×⅛	110	1,424.47 kg	−488.79 kg

All retain six HSS 12×4×¼ bolster segments, two continuous main girders, two continuous perimeter rails and four lines of HSS 3×1×⅛ seat rails. Secondary-member changes include the perimeter rails. No body stiffness is credited. These are material-accounting alternatives, not interchangeable approved parts.

Use D1 as the lower-change comparison and **D2 as the focus of the next floor/frame engineering iteration**. D3 remains a stiffness and mass sensitivity. D2 is not selected for construction: even its bare steel exceeds 1,500 kg by 27.29 kg, before required hardware. D3 leaves only 75.53 kg before those omissions. An explicitly hypothetical 150 kg of omitted parts would produce 1,677.29 kg for D2 and 1,574.47 kg for D3; this is reserve arithmetic, not a hardware estimate.

2. Geometry and support changes are explicit

Coordinates are inches, x0 at the front inside wall, y0 on the centreline and z0 at the rail. Keep frame x−47…353, body x−3…309, width y±55, main-girder centrelines y±34 and bogies at x45/261. Every alternative has a complete, disjoint member-coordinate list in structure.json; no predecessor file is overwritten.

D2 and D3 cabin nodes are **−3,21,45,69,93,117,141,165,189,213,237,261,285,309**. Platform nodes remain −46,−25,331,352. At x45 and261 the 12-inch-deep bolster segments replace ordinary crossmembers. Transverse segments run y−53…−36, −32…32 and36…53; their ends meet the main/perimeter faces without overlapping their modeled volumes. That meeting is not an engineered connection.

All existing transverse panel seams at x45,93,141,189,237,285 remain supported. The stock-panel layout's longitudinal seams remain over the mains at y±34. A 24-inch ordinary bay has 22 inches between nominal member faces; bays adjacent to the four-inch-wide bolster have 21 inches. Calculations use the longer centre spacing. Actual supported width, joint gap, panel closure and edge fastening are still return fields.

Four seat-rail lines remain at y−44,−20,20,44. Longer segments add **7.38 kg** of seat rail compared with D1; removing crossmember segments saves **140.76 kg**, giving the net 133.39 kg D1→D2 reduction. This accounting captures the support members that remain necessary. The 110-member D2 total comprises two mains, six bolster segments, 48 ordinary transverse segments, two perimeter rails and 52 seat-rail segments.

The cabin dimensions and 32 places are retained. Eight seat rows remain x41,73,...265. No rail is removed beneath a seat anchor to make the mass result smaller.

3. Properties, loads and new calculations

The manufacturer-hosted Atlas table was rechecked on 18 October 2026. The five study sections use its strong-axis I/S and nominal mass C3DS01:

HSS, inches	I, in⁴	S, in³	lb/ft
12×4×¼	119	19.8	25.82
10×4×¼	74.7	14.9	22.42
4×2×3/16	3.66	1.83	6.87
4×2×⅛	2.65	1.33	4.75
3×1×⅛	0.817	0.545	3.05

Nominal walls ¼, 3/16 and ⅛ correspond to table design walls 0.233, 0.174 and0.116 inch. Current access does not establish stock, grade, toughness or application approval. Display models use sharp-corner nominal hollow shapes; material mass is calculated from the table, not those approximate volumes.

The portable structure.py now carries the actual assumed seat loads through **feet → longitudinal seat rails → transverse nodes → main girders → bogie support stations**. Each two-person bench has 290 kg passengers plus a 15 kg assumed bench mass. Four study feet at row-centre±6 inches and the relevant y20/44 lines each apply 747.76 N. The actual seat and anchor arrangement remains unknown.

For each rail bay, the script combines every foot within that bay, calculates its two support reactions, and passes those reactions into the transverse beam. It does not assess a single isolated foot while ignoring a second foot on the same span. Total transferred bench load equals all 16 occupied benches. Rail reactions use transverse-node centrelines; physical tubes stop at member faces. Connection offsets and feet near those faces require local joint/anchor analysis.

Local static sensitivity	D0	D1	D2 / D3
Largest seat-derived transverse-node load	3.739 kN	3.739 kN	4.736 kN
Largest seat-rail bending stress	7.97 MPa	7.97 MPa	12.76 MPa
Largest seat-rail elastic deflection	0.0141 mm	0.0141 mm	0.0713 mm
4 kPa floor trial: crossmember stress	13.22 MPa	18.19 MPa	27.28 MPa
4 kPa floor trial: crossmember deflection	0.362 mm	0.499 mm	0.749 mm

The 4 kPa trial is an analyst sensitivity, not an adopted code pressure, and is not added to the full seated-payload case. Platform members retain a separate two-adult outer-edge case: thinner members produce 66.29 MPa and2.479 mm, with a −0.836 kN opposite reaction. The assumed continuity therefore needs an actual restraint/moment-transfer detail; these beam results do not approve thinner walls or predict whole-coach overturning.

For a common global comparison, 16 occupied benches total4,880 kg, a crew member at x153 contributes145 kg, and the remaining **8,175 kg** is uniformly distributed over the400-inch frame. Each main receives the calculated transverse reactions plus half the centred crew and residual loads. Sum=13,200 kg in every alternative; lighter members do not reduce the comparison demand.

Main-girder static screen	D0 / D1	D2	D3
Seat-transfer case: maximum bending stress	56.71 MPa	56.53 MPa	75.12 MPa

Main-girder static screen	D0 / D1	D2	D3
Seat-transfer case: maximum deflection	5.059 mm	5.035 mm	8.020 mm
Alternative full-gross cabin UDL: maximum stress	76.00 MPa	76.00 MPa	101.00 MPa
Alternative cabin UDL: maximum displacement	8.823 mm	8.823 mm	14.055 mm

The seat-transfer maximum displacement is at midspan; the cabin-UDL case's maximum is at a free platform end. Each main's ideal symmetric support reaction is32.362 kN at each bogie station. This does **not** replace the retained asymmetric front-bogie81.343 kN/32.578 kNm roll case or other combinations. Actual equipment, body and bogie mass locations must replace the residual UDL.

These are unfactored Euler-Bernoulli screens using E=200 GPa, not stress acceptance or fatigue checks. The program verifies force balance, support deflections, free-end moment, benchmark mass, zero positive-volume member intersections, seam support, and independent point-load/overhang closed forms. Shear distortion, joint flexibility, local wall failure and 3D torsion are not modeled.

4. Floor and bogie interfaces govern the choice

The unchanged 25.30 mm study floor stack places steel at z27.00394 inches and every retained 12-inch bolster underside atz15.00394. D3 raises only the 10-inch main underside toz17.00394; it does not raise the bogie/bolster interface. A thicker floor panel with finished height fixed lowers the steel and bolster further, as the body study separately calculates.

The required condition remains bogie support-top height + attachment stack + relative-motion clearance ≤ bolster underside height, with all three supplier terms unresolved. Do not notch members to force a fit. D3 also creates a two-inch depth step at the main/bolster junction requiring an explicit joint design.

Changing from16 to24 inches increases bending-only panel deflection by5.0625 at the same load and rigidity. The body module compares0.75/1.0-inch Plascore options and published-property discrepancies separately. A supported seam alone does not qualify the floor: core shear, wheel/heel indentation, rolling fatigue, edges, inserts and attachment compatibility remain necessary C3DS03.

5. Release sequence

First resolve a complete panel construction and span with the supplier and engineer. Then obtain the actual bogie/support envelope and load-sharing scheme. Carry D1/D2 into a full frame comparison with actual component locations; retain D3 only if stiffness, fatigue and connection results justify its additional saving.

The responsible structural engineer must define combined vehicle loads, analyze local HSS walls and joints, and release material grades, tolerances, welds, fasteners and inspection criteria. HSS member checks cannot substitute for combined-force connection checks C3DS02. Thinner walls require specific fabrication and durability review.

Close the omitted draft structure, reinforcement, mounts, seat hard points, stairs/guards, coating and lifting/jacking masses before revising the vehicle ledger. Accessible lift and wheelchair-restraint anchors need a distinct design. Farm crews can develop full-size mockups and approved fixtures; this issue authorizes neither structural cutting nor passenger loading.

04 | Running gear, floor height and wheel-rail interfaces

Keep a complete, engineered, braked two-axle bogie under each coach end. The research has not established an off-the-shelf bogie that fits this car. This phase makes the fit test explicit: a real supplier assembly must satisfy the load cases, wheel/rail interface and moving clearances together. The **28-inch floor remains an unresolved target**, with 32/36-inch floors compared only as alternatives. The owner's 0% longitudinal grade is retained; it does not eliminate curvature, track twist, boarding imbalance or braking loads.

1. Actual products and responsible supply routes

Primary pages were checked **2026-10-18 UTC**. No quotation, project approval or supplier response was obtained. Public catalog presence is not confirmed availability.

Route / exact reference	Verified public evidence	Engineering disposition
Western Train custom coach running gear	36-inch offering, custom parts and drive-truck regauging; coach air brakes C3DG01	Preferred complete-OEM inquiry route. Standalone unpowered bogie supply, load ratings, geometry and hydraulic compatibility remain unverified.
Severn Lamb **Jupiter**	914 mm gauge, 50 m minimum radius, 16 km/h maximum, enclosed-coach option C3DG02	Identified whole-train reference only. Its specifications do not rate our train. Confirm 914.4 mm interface and standalone supply.
Railteco **RTKZ1** forest-railway passenger bogie	762 mm gauge, 596 mm wheels, 1,165 mm wheelbase, ≤1.5 t tare, published 3.75 t maximum axle; centre plate and sprung sidebearers C3DG03 C3DG04	**Reject as supplied:** different gauge and inadequate axle-sum screen for the severe regular-coach case. It is a useful geometric reference, not a regauging recommendation.
CFD two-stage passenger bogie	2.4 m wheelbase, 5-8 t bogie mass; 1000/(1067)/1435 mm gauge family C3DG05	**Reject baseline:** two bogies alone weigh 10-16 t, above the entire 7,040 kg regular tare allocation.
UTCRAS / GHH-BONATRANS	Wheelset engineering, assembly/testing and custom wheelsets C3DG06 C3DG07	Specialist alternative under a named bogie design lead. Neither capability page supplies a compatible part number.

Railteco's corporate site links its sales site, supporting manufacturer provenance. The sales page has "762m" and "RTKZ32" copy errors within the RTKZ1 listing; use a controlled drawing before procurement. Generic certification claims on that page are not project approvals. Do not adopt its advertised delivery wording as a current lead time.

2. The floor-height inequality now has numerical consequences

Retain the Phase 3C trial floor stack: 25.30 mm including the 19.05 mm panel, 2.25 mm finish and unselected adhesive/bedding allowances. The necessary central-bolster condition is:

Hsupport + tattachment + crelative-motion ≤ Hfloor −25.30 mm −304.80 mm.

Every actual supplier term on the left is **null** in the register. A positive result would be only a necessary local condition, not proof that wheels, brakes, suspension and the entire frame clear.

Finished floor	Steel bearing plane	12-inch central-bolster underside	4-inch crossmember underside
28 in retained target	685.9 mm /27.00394 in	381.1 mm /15.00394 in	584.3 mm
32 in comparison	787.5 mm /31.00394 in	482.7 mm /19.00394 in	685.9 mm
36 in comparison	889.1 mm /35.00394 in	584.3 mm /23.00394 in	787.5 mm

With **trial** attachment and movement allowances of 25.4 mm each, the 28-inch scheme admits a support top no higher than **330.3 mm**. A hypothetical 400 mm support needs at least 30.744 inches of finished-floor height; 450 mm needs 32.713 inches; 500 mm needs 34.681 inches. These are packaging arithmetic, not measured bogies, selected clearances or permission to raise the floor. A 10-inch main girder does not remove the retained 12-inch centre-bolster constraint. Do not notch it to fit an unknown truck.

Wheel diameter is a separate test. The executable cylinder-envelope study uses 400/500/596 mm diameters, 1000/1165/1500 mm wheelbases, a clearly assumed 101.6 mm wheel-envelope width, and wheel centre planes at ±457.2 mm as a **contact-location proxy**, not an approved profile. It checks both bogies against the ordinary crossmember strips of all four D0-D3 alternatives and yaw extremes. At 1,165 mm wheelbase, the 596 mm trial intersects rear crossmember x237 by **11.7 mm**; a 25.4 mm relative upward-motion trial increases interference to **37.1 mm**. The 500 mm trial instead leaves 58.9 mm under that motion assumption. None is a complete clearance acceptance: actual tread/flange shapes, axleboxes, brakes, suspension motion, wear and tolerances remain absent. The useful result is that a wheel can fit below the floor panel yet hit a crossmember.

3. Curving and swept-space requirements

The calculator uses exact rigid chords on a common circular centreline. With curve radius R, bogie wheelbase w and 216-inch bogie spacing L:

$Rb=\sqrt{(R^2-(w/2)^2)}$; $Rc=\sqrt{(Rb^2-(L/2)^2)}$; relative yaw$=\operatorname{asin}[L/(2Rb)]$.

For the 1,165 mm reference wheelbase:

Curve radius trial	Bogie/body yaw	Body-centre inward offset	Frame envelope: outer / inner from track centreline
30 m	5.247°	131.4 mm	1675.6 /1528.4 mm
50 m	3.145°	78.7 mm	1569.1 /1475.7 mm
75 m	2.096°	52.4 mm	1513.4 /1449.4 mm

Frame values include the full 400×110-inch rectangle, including platforms. They exclude stairs, handrails, body roll, lateral play and required clearance margins. They are not a site clearance line or approved radius. Reverse curves and transitions require the actual connected-train kinematic solution; one-circle results cannot authorize back-to-back curves.

The JSON contains 81 rectangular bogie-envelope sweeps with widths 1200/1500/1800 mm and lengths wheelbase+wheel diameter. These are deliberately labeled parametric boxes, not invented bogie frames. For half-length a, half-width b and yaw θ, the body-axis half-width is $a\sin|\theta|+b\cos\theta$. Compare it with the 812.8 mm main-girder inner-face coordinate only where their vertical bands overlap. The supplier must replace the box with new/worn, empty/full, bump/rebound, yaw, roll, pitch and lateral-travel CAD envelopes.

4. Axle loads and roll support cannot be chosen by mass divided by four

The retained 13,200/11,000 kg car values are upper analysis cases, not ratings. The regular front-half crowd case produces **8,294.69 kg-equivalent gross reaction** at one bogie under the retained symmetric, centred bogie-mass idealization. The 13,200 kg vehicle total already includes bogies; do not add bogie mass again to that axle-sum reaction. Even perfect axle sharing is 4,147.34 kg per axle; no sharing arrangement can keep both below the RTKZ1's combined published 7,500 kg two-axle limit. The static sum exceeds that screen by 794.69 kg. Its 762 mm gauge independently prevents adoption.

For an actual bogie, $N_{front/rear} = R_{bogie}/2 \pm M_{pitch}/w$. The 36-case sensitivity matrix varies wheelbase and an explicitly unselected 0/2/4 kNm bogie pitch moment. At 1,165 mm, 4 kNm shifts **350.12 kg-equivalent** between axles, raising the regular front-crowd high-axle screen to 4,497.46 kg. This moment is not added silently to an approved load case. The existing separate 0.8 m/s², assumed 1.5 m CG braking sensitivity raises the front bogie to 8,589.09 kg-equivalent; real coupler heights and suspension reactions must replace that idealization. Separate sprung and unsprung loads and include fatigue, track twist, dynamics and individual wheel forces in the supplier analysis.

The inherited **81.343 kN gross-equivalent** front reaction must be separated from body-supported load before sizing sidebearers. The revised 36-case sweep subtracts assumed bogie masses of 0/750/1500 kg; actual bogie mass is null. Retain the 32.578 kNm crowd-induced roll demand only under the centred bogie-tare assumption. For supported load Ns and centre share p, a compression-only pair requires $s \geq 2T/[(1-p)Ns]$. At the **1500 kg trial**, supported load is66.633 kN; minimum spacings become **977.8/1303.8/1955.7 mm** for p=0/0.25/0.50. At 36-inch spacing even p=0 gives **−2.312 kN** on one side. This demands a different support model or designed tension/retention path; it is not a selected layout. The zero-bogie-mass case reproduces Phase3C for comparison, but overstates available compression if treated as the real supported load. Frame strength alone does not establish individual wheel unloading or stability.

5. Real rail couplers beneath the platforms

Two documented rail-specific routes warrant dimensional returns. Voith's **Type430/530** family is compact/folding and lists tension/compression strength up to 300 kN C3DG08. Dellner's **people-mover semi-permanent coupler** uses a double-acting damper, pivot anchor and cable support C3DG09. Neither page establishes a complete assembly that fits this train; family strength is not a working/fatigue rating. Semi-permanent connections merit study between coaches that are rarely separated; the locomotive interface may need a different operational solution.

The retained 30-inch gap and 14-inch setback give 58 inches between study pivots. At the front, pivot x−33 and frame edge x−47 leave a 14-inch concealed length. An independent joint-angle trial of 5/10/15° moves the connection axis laterally **1.225/2.469/3.751 inches** at that edge, before hardware width and movement allowance. These angles are not the bogie yaw values. The x−46 crossmember occupies this route; its actual vertical relation to the selected assembly must be designed. The JSON leaves required opening size null until the hardware envelope is returned.

Require combined tension/compression, shear and vertical ratings; slack/stroke and energy curves; articulation, retention, wear limits, maintenance access and mounting reactions into **both main girders**. Resolve reversing and emergency brake imbalance using the coordinated brake cases. Keep the coupling and cables guarded from passengers while preserving inspection access. No pin, plate, weld or ordinary trailer component is selected.

6. Release the wheel and track interfaces together

The track engineer and bogie supplier must agree one dimensional record: new/worn wheel profile coordinates and measurement datums, back-to-back, flange thickness/height, rim width, diameter range, rail profile/grade/inclination, gauge measurement plane/limits, guard/check gauge, frog and crossing flangeways, curvature, twist and wear limits. A common nominal gauge is insufficient. Atlantic's 40/60 ASCE tables identify candidate rail sections C3DG10; they do not establish wheel contact or track capacity.

Issue the twelve open interface returns with the load and fit calculations. Obtain two complete bogies for the first prototype only after responsibility, drawings and accepted tests are defined; they form part of the eventual eight-bogie lot. Farm work can advance the cabin mockup, survey and approved track preparation.

Wheel machining, axle fits, bogie/frame fabrication and qualification remain specialist work. Choose the floor height only after the real package passes these coordinated checks, then revise stairs, accessible boarding, stations and mass/CG together.

05 | Floor/body performance and complete-mass reconciliation

PF-TRAIN-S3D-R0 · 18 October 2026 UTC · Preliminary development; not fabrication release. Retain the 306 × 104 in clear cabin, 32-place regular seating trial, accessible layout and 28 in finished-floor datum. The lighter D2 chassis is a study with 24 in crossmember spacing. Keeping the existing panel material on that wider grid requires new evidence; a steel weight reduction does not establish a complete lighter coach.

Floor response: measured properties now inform the comparison

Plascore publishes AA3.6-80 rigidity for 3/4 and 1 in panels, but its webpage and marine brochure disagree. The webpage gives 46,858/71,654; the brochure gives 44,000/71,000 **lb·in² per inch of width**. The brochure image confirms the units. Both reference four-point, quarter-point testing at 20 in span and 3 in specimen width. Published data do not establish the core direction, minimum production values or separated bending and shear stiffness. The webpage's SI heading is unsuitable for a reliable conversion, so this screen uses the imperial column. [C3DB01-02]

Use the lower reported values for an expressly limited Euler beam comparison: a one-inch-wide simply supported strip, uniform 4.8 kPa pressure, and deflection proxy $5qL^4/(384D)$. The pressure and L/360 comparison objective are analyst assumptions for the engineering discussion, not adopted vehicle-code requirements. Substituting test-derived rigidity does not establish a true bending-only stiffness; core shear, local indentation, joint behavior, fatigue and creep remain outside this model. These results are neither a bound on actual deflection nor a strength pass.

Panel	Support centers	Deflection proxy	Analyst L/360 objective
3/4 in	16 in	0.343 mm	1.129 mm
3/4 in	24 in	1.736 mm	1.693 mm
1 in	16 in	0.213 mm	1.129 mm
1 in	24 in	1.076 mm	1.693 mm

At equal material and load, changing 16 to 24 in multiplies this proxy by **5.0625**. Even the simplified 3/4 in case slightly exceeds the chosen comparison objective. Retain D2 as the next engineering/test candidate, not a released floor arrangement. The 1 in candidate merits testing, but its favorable uniform-load number does not establish caster, seat or doorway strength.

The calculation file also sweeps occupied-device masses 295/445 kg, wheel shares 25/50%, and vertical multipliers 1/1.5. These are demand sensitivities, not admitted-device limits. A center-load strip study varies effective load-spreading width from 6 to 24 in; that width is deliberately unproved. Actual wheel patch, orientation, load transfer at seams and local skin indentation must come from device data and panel testing. Do not use a uniform pressure calculation to approve concentrated loads.

Keep floor elevation and supports coordinated

Canonical F075 remains 19.05 mm panel + 2.25 mm Sirius NT finish + 1 mm adhesive reservation + 3 mm bedding/isolation reservation: **25.30 mm total**, steel bearing z = **27.003937 in**. The last two layers remain packaging reservations, not specified products or installation thicknesses. Support bedding is local to supports. [C3DB03]

F100, the separate 1 in panel alternative, totals 31.65 mm and adds **7.57 kg** across the cabin. To preserve the finished floor, it lowers steel bearing to **26.753937 in** and the retained 12 in center bolster underside to **14.753937 in**. The canonical drawings/model retain F075. Bogie support height, attachment stack and suspension clearance must close before selecting F100.

All existing transverse panel seams remain at x = −3, 45, 93, 141, 189, 237, 285 and 309 in; every seam coincides with the D2 grid. Stock-sheet longitudinal seams stay at y = ±34 in over main girders. Whole-width and supported 4 × 8 sheet studies remain available; actual edge gaps, closures, bearing and kerf still prevent a production cut release.

D1-to-D2 bare-steel reduction is about **133.39 kg**. A simultaneous F075-to-F100 change consumes 7.57 kg, leaving **125.82 kg before any added hard-point or panel-support reinforcement**. That added delta is unknown. Weigh it rather than treating the full steel reduction as coach savings. These comparisons do not alter the retained 7,040/7,535 kg tare allocations.

Hard points must reach the frame

Seat-row reference stations remain x = 41 + 32i, i = 0…7. Candidate anchor rails remain y = ±20 and ±44 in. The structure module's feet at row ±6 in are a load-distribution study; no production hole pattern follows from them. Actual bench supplier footprints and occupied mockup results govern. Carry seat vertical, braking, lateral and fatigue loads through engineered sleeves/inserts/backing into the rails and their frame joints. Compressible bedding, finish and thin skins cannot serve as the assumed bolt-clamp stack. [C3DB04]

Wheelchair bays remain x = 60…120 and 120…180, y = −52…−16 in. These are clear-space reservations, not tie-down anchor coordinates. Restraint geometry, floor-anchor forces and any shoulder-belt loads need selected hardware and a direct load path into the frame/body structure. Keep covers removable for anchor inspection.

Normal station bridge boarding remains the space-efficient basis. The optional lift's conservative reservation x = 4.52…55.48, y = 34…52 in is not actual CAD or a bolt pattern. Braun's retained reference is 385 lb / 174.63 kg; mounting reinforcements, tower loads, power system and bridge interface add mass. Motor-vehicle mounting kits cannot be assumed to fit this rail frame. Preserve both the optional lift and independent rescue holds. [C3DB11]

Mass ledger: partial materials, allocations and missing details

The ledger now covers every retained coach subsystem and separates typical material takeoffs, conditional material studies, whole-subsystem allowances and unquantified details. Blank masses are **unknown, not zero**. Subsystem allowances include their detailed items; they must not be added to those items again. Growth remains included once.

Partial takeoff or remaining target	Regular	Accessible
Wall, roof and ceiling panels	445.86 kg	424.62 kg
Floor panel plus finish	146.79 kg	146.79 kg
Bare glass, 6 mm total glass-ply study	129.44 kg	100.03 kg
Remaining body target after panels and that glass study	324.70 kg	425.35 kg
Remaining floor/insulation/finish target	253.21 kg	273.21 kg

The regular takeoff uses sixteen 22 × 38 in window openings and two 32 × 78 in end doors. The accessible takeoff retains four 35 × 38 and six 22 × 38 windows, two 36 × 78 end doors and two 42 × 78 side doors. Openings are deducted from both liner and outer panel coverage. Roof and ceiling remain flat proxies; crown, overlaps and weather cap add material.

Glass density is documented at 2,500 kg/m³. The 6/8/12 mm total glass-ply sensitivities are mass comparisons, not selected tempered/laminated/insulated glazing or emergency-exit specifications. Frames, seals, interlayers and glass extending behind openings remain unknown. [C3DB05]

The remaining body target must cover posts, headers, bows, structural mounts, complete doors, glazing retainers, roof weathering and trim. No defensible member sections or complete-door masses exist yet. The apparent headroom is not savings. Insulation volume is calculable, but the reviewed Armacell page supplies no density; retain its mass as null. Adhesive mass requires installed joint volume, even where product density is published. [C3DB06,08]

Coordinate the mass references with the utility module: Konvekta HVAC7403 at 520 kg leaves only 30 kg within the regular 550 kg HVAC/duct allowance; HVAC5401 at 275 kg leaves 175 kg within the accessible 450 kg allowance. Both exclude project mounts and ducts. If the larger 7403 were instead used on the accessible coach, it alone would exceed that allowance by 70 kg; this is a separate sensitivity. These are unselected old-catalog references; current equipment and ambient performance must be established. Structural roof supports belong in the body ledger while equipment/ducts belong in HVAC, avoiding duplicate counting. [C3DB07]

Thermal, fire and durability evidence

At 30°C mean temperature, the published ArmaFlex expression gives conductivity 0.04381 W/(m·K); a 13 mm layer has R ≈ 0.297 m²K/W. This layer alone does not substantiate the earlier opaque-U assumption of 0.7 W/(m²K). Calculate the complete wall, floor and roof, including metallic bridges, air films, openings, ducts and condensation conditions. The 57.15 mm cavity also contains framing and services. Do not fill it blindly with insulation or extend the product's 3-13 mm fire claim to a thicker composite assembly. [C3DB06]

The body engineer must agree the applicable fire/smoke and evacuation basis, then qualify the actual panel/adhesive/liner/insulation/finish combination. A material listing is not complete-coach compliance. Preserve drainage, sealed honeycomb edges, dissimilar-metal isolation and intentional electrical bonding. Test actual coatings, cleaners, water exposure and thermal movement; Sika's typical lap-shear number is not an allowable joint stress. [C3DB08]

Prototype sequence and release evidence

Use seven recorded test packages: directional material coupons; a full-width support-bay/seam assembly; a seat hard point; wheelchair/lift structural interfaces; a window/door/roof corner; floor rolling/cleaning durability; and final module/car weighing. The register identifies the purpose, evidence and held acceptance fields for each. The laboratory and responsible engineer select test editions, specimens, load combinations, cycles and acceptance values before loading. ASTM public scopes identify methods for separating sandwich flexural/shear properties; full procedures and application acceptance remain to be obtained. [C3DB09-10]

Farm crew can make nonstructural templates, label and weigh samples, record dimensions and assist with controlled leak/cleaning observations. Specialists design/load structural tests and release critical anchors, bonding and lift/roof supports. Use ballast or fixtures, not people, for proof loads. Do not cut production panels or fabricate critical attachments until those specific interfaces are released.

06 | Braking, hydraulic response and electrical interfaces

Recommended development arrangement

Retain hydraulic service braking, with one controlled zone per bogie and local stored mechanical application energy. Develop spring-applied, hydraulically released actuators only with a supplier that accepts repeated service modulation, unloaded braking and the complete failure response. If that cannot be supported, compare active hydraulic service calipers plus a separately engineered emergency/parking system against a complete automatic-air package. Hydraulic propulsion and hydraulic brakes remain distinct choices.

Prefer a matched brake supplier package. Knorr-Bremse HSC1P10N and Hanning & Kahl units are documented candidates, **not a verified compatible assembly**. HSC1P10N lists 20 kN per disc, two release circuits, 45 mm maximum disc width and approximately 15 kg; the force convention, pressure and mounting details remain unverified. [C3DP01] Hanning's EHM lists two-bogie control, 24 VDC power, a 48 VDC control wire and 160 bar permissible pressure. That pressure is not the selected release pressure. [C3DP02] The configurable ECU/HCU route requires current, flow, force and failure documentation; the ECU's announced 2028 transition must be reflected in the offered configuration. [C3DP03-04]

Wheel torque and mounting loads

Use the retained **65,100 kg upper engineering case** and **0% longitudinal grade**. At the assumed 0.50/0.80 m/s² service/emergency targets, total force is 32.55/52.08 kN. Equal four-axle allocation gives 2.64 kN per regular-coach axle and 2.20 kN per accessible-coach axle in emergency.

Wheel diameter sensitivity	Regular upper emergency torque per axle	Accessible upper emergency torque per axle
400 mm, analyst trial	528.00 N·m	440.00 N·m
500 mm, analyst trial	660.00 N·m	550.00 N·m
596 mm, catalog reference only	786.72 N·m	655.60 N·m

The running-gear register identifies the 596 mm Railteco RTKZ1 reference as a **762 mm gauge tread-braked bogie**, incompatible as published. It does not establish a disc package or a selected wheel. The calculator retains one/two disc-per-axle alternatives separately.

For each disc, use torque divided by effective disc radius to obtain tangential mount reaction. Required normal force summed across both friction faces is **torque / (pad friction × effective radius)**. At 660 N·m, one disc, assumed friction 0.25 and effective radius 0.12 m, the result is 22.0 kN total normal force, 11.0 kN per equal face and 5.50 kN tangential reaction. These assumed values cannot be compared directly with the catalog's undefined 20 kN convention. Add actual bracket offsets, caliper inertia, disc/hub loads, vibration, wear and fatigue before designing attachments.

Equal-load ideal adhesion is 0.0816 for 0.80 m/s² with all axles braked. Concentrating the same demand on two of four equally loaded axles doubles it to 0.1632. These are lower-bound screens without unloading, curves or contamination. A fixed 10.56 kN brake force on a 7040 kg empty regular coach produces 1.50 m/s² and ideal adhesion 0.153; empty/loaded force control is therefore a release requirement. No operation with an isolated bogie is authorized.

Response, fluid displacement and energy

A new piecewise analytical integration independently reproduces the Phase 3C response screens. At 7.5 mph, a 13,200 kg regular coach has 74.19 kJ translational energy: 18.55 kJ per disc with four equal discs, before rotating inertia or unequal sharing. This is an energy input, not a temperature/fade rating. At 7.5 mph, the assumed 3 s step-delay emergency stop remains **17.084 m**. Assumed 0.5 s delay plus 1.5 s linear buildup gives 11.142 m. Delays staggered from 0.3-1.5 s give 12.304 m; adding 5% equivalent rotating inertia gives 12.662 m. Under the synchronous profile, losing half of one regular coach's brake force gives 11.942 m and losing the whole coach gives 12.944 m. These are ideal simulations, not acceptance distances; adhesion, slack impact and fade are excluded.

Obtain the actual fluid displacement versus travel/pressure, hose compliance, leakage and drain-flow curves. The average-flow screen is **60 × effective volume in litres / response seconds**, plus leakage. For illustration only, a zone displacing 0.10 L in 0.50 s requires 12 L/min average; this is not a selected pump or actuator rating. Cold viscosity, pressure decay, valve opening and line losses determine the actual force history.

The calculator also normalizes pressure work to 100 bar: 0.10 L corresponds to 1 kJ for a constant-pressure displacement. **This is neither spring energy nor available braking energy.** Spring energy requires the actual force-travel integral; stored hydraulic pressure can hold spring brakes released. Demonstrate application with power loss, blocked return, trapped auxiliary pressure and the agreed valve faults. A second auxiliary circuit releases the brake and must be monitored and controlled during recovery. [C3DP05]

Functional electrical arrangement

Develop either shore power or a diesel hotel generator through interlocked source selection into protected coach distribution. Each coach receives a local isolator, matched HVAC/lighting distribution and a separately protected charger. Brake power is local, nominal 24 V for this study, segregated from hotel loads; the EHM's 48 V control-wire requirement cannot be replaced without OEM agreement. Provide separate emergency-light reserve and fault containment so an HVAC, charger or passenger-control fault cannot defeat brake application.

Service demand enters the brake ECU. A separately monitored emergency/separation function directly requires application at both bogies and removes locomotive traction permission; it must not depend solely on the ordinary service bus. The functional drawing intentionally omits valve arrangements, connector pinouts, wire gauges and fuse settings. No intercar hydraulic service hose is proposed. Electrical continuity alone cannot detect a broken coupler while jumpers remain attached.

On hotel blackout, stop HVAC, retain emergency lighting and execute the approved operational stop without losing brake functions. On brake DC loss, local application and traction inhibition must occur without requiring surviving hotel AC. Source return or controller reboot must not automatically release parked cars. Shore connection must inhibit movement. The complete bonding/neutral arrangement, protection and disconnect sequence remain electrical-engineer decisions; running rails and frames are not assumed normal return conductors.

HVAC and generator operating envelope

The retained cooling screens are 22.70 kW per regular coach and 13.28 kW accessible, totaling **81.38 kW thermal**, at 110°F outdoor/75°F indoor with the prior ventilation/door assumptions. The new body study shows that its 13 mm insulation layer does not substantiate the previous opaque-U assumption; recalculate cooling and heating from the completed envelope, thermal bridges and ventilation before selecting equipment. These retained loads are comparison scenarios, not demonstrated requirements. Current public documentation was checked on 18 October 2026; it does not supply matched operating currents at those conditions.

Konvekta's August 2021 catalog lists 7403/5401 at 400 VAC with 24 VDC controls, cooling ratings of 30/16 kW at different stated conditions and heaters of 30/7.2 kW. Its rating conditions, frequency acceptance, compressor current, start current, minimum circuit ampacity and maximum protection still need a controlled current offer. Do not connect catalog 400 V equipment directly to the 480 V study. [C3DP06] Northwest Rail Electric's NW-64-CN normally uses 480 V three phase, but its compressor and matched evaporator vary; no

project electrical load is inferred from nominal tons. [C3DP07]

Cooling sensitivity, assumed whole-system COP	Electrical load including assumed 4 kW auxiliary allowance	Balanced current at 480 V three phase, assumed PF 0.90
2.0	44.69 kW	59.73 A
2.5	36.55 kW	48.85 A
3.0	31.13 kW	41.60 A

The auxiliary allowance is not an equipment takeoff. The published heating capacities sum to 97.2 kW thermal. **If supplied entirely by electric resistance heat at assumed unity conversion**, they imply a 97.2 kW heater input and **101.2 kW** illustrative total with that allowance. The 5401 page explicitly identifies electric heating; the 7403 table identifies a 400 VAC heating supply without a complete electrical input schedule. Treat 101.2 kW as a resistance-only scenario, not measured or manufacturer-rated bus demand. Confirm heater technology, actual winter loads and staging before generation selection; this resistance-only scenario exceeds either previous generator size.

Stadco's 6-42T4F/6-80T4 reference continuous outputs are 38/70 kW. Its T240/T480 alternator starting figures are 124/270 starting kVA at **35% voltage dip**, not a project-approved dip or a complete genset transient guarantee. Ratings use 80°F, 500 ft and 50% RH; obtain 110°F/site derating and the actual engine/alternator configuration. [C3DP09] The 38 kW reference has only 1.45 kW margin against the COP 2.5 illustrative cooling load before derating and omitted loads. Neither generator is selected.

Starting is a separate test: the calculator evaluates assumed 5/10 kW compressor stages and 3×/5× current, with remaining loads already running. These multipliers are not manufacturer data. Require actual start current, duration, restart delay, power factor and VFD behavior; test sequential restart, voltage/frequency recovery and simultaneous restart after blackout.

The NW-25100 is a useful custom utility-locker lead with staged HVAC, lower-voltage distribution and emergency-light options; it is not a safety brake controller. [C3DP08] Full-sized HEP examples establish no project conductor sizes. [C3DP10] Cable design must use each segment's downstream loads, route length, flexible service, ambient/bundling, voltage drop, fault current, short-circuit withstand, protection coordination, connector ratings and bond continuity.

Evidence needed before installation

The saved register contains ten interface returns and ten unperformed test records. Obtain controlled brake/wheel/disc CAD and force curves first; then complete guarded actuator tests, secured coach fault tests and unoccupied contained braking trials under engineer-approved methods and limits. Record actual wheel loads, independent ground speed, axle speed, pressure, force, temperature, voltage and synchronized timestamps. Electrical trials must include phase errors, source isolation, ground-fault protection, HVAC starting and blackout reserve. Moving breakaway tests require a separate engineered containment plan.

Farm crew may support labeling, protected routing and records under released drawings. Specialists must specify and install brake hydraulics, controls, high-voltage distribution, protective settings and the acceptance instrumentation. No selected caliper mount, conductor, pressure setting, stock confirmation or supplier approval is implied.

07 | Engineering handoff and first physical work

Order	Work and responsible lead	Evidence before proceeding
1	Owner/farm crew: stable ground-level occupied mockup using the retained interior alternatives. Record seated/standing adults, doors, winter clothing and accessible-device circulation.	Measured observations and photographs; no claim that a drawing alone validates comfort or evacuation.
2	Vehicle engineer and selected suppliers: exchange the structured wheel/bogie, brake, coupling, panel, body and electrical returns.	Exact configured drawings, operating envelopes, scope exclusions and responsibility for every interface.
3	Vehicle/structural engineer: select the frame/floor combination and close the complete mass and centre-of-gravity budget.	Adopted load/fatigue basis; joint, local HSS, deflection, wheel unloading and stability assessment; accepted dimensions and tolerances.
4	Specialist fabricator and engineer: develop production joints, welds, fastening, lifting/jig design and inspection points using the retained Phase3C traveler.	Released shop documents, qualified procedures, material certificates, accepted inspection/testing criteria and controlled deviations.
5	Body and brake/electrical integrators: qualify panel/hardpoint interfaces and configured equipment; assemble the regular first article.	Recorded panel/joint/attachment evidence, correct torque and clearances, hydraulic/electrical test results, measured mass and reactions.
6	Responsible engineers and operator: approve staged stationary and moving trials before duplication or passengers.	An accepted test route, restrained ballast, instruments, numeric limits, fault cases, evacuation plan and written release records.

The accessible coach requires its own structural, boarding and evacuation verification. A successful regular prototype does not automatically release the accessible vehicle. The farm crew can prepare mockups and survey/control records, and undertake only the fabrication or track work assigned by the reviewed installation/fabrication plan.

08 | Formats, scope and verification

Read the report alongside the Phase3D component-integration drawings and editable workbook. The rotating models and STEP exports show the four frame alternatives and supported panel envelopes. The earlier complete-coach space studies and six-adult interior comparisons remain on the same site.

Format	Purpose and limitation
PDF / SVG	Readable engineering review sheets. Critical unselected hardware and connections are marked as held.
DXF	Sheet DXFs use paper inches; separately named full-size frame alternatives use model inches. Consult the drawing README before using coordinates.
STEP / web meshes	STEP solids use millimetres; browser geometry uses inches. Nominal hollow-section envelopes omit real corner radii, weld preparations and connections. No native CAD feature tree or manufacturing release is included.
XLSX / JSON / scripts	Editable studies, source records and interface returns. Blank evidence/rating fields mean unresolved. Formulas and exported values are checked; they do not establish that the assumptions are accepted.
Source package	Portable engineering inputs, generators, calculation results and review records. Manufacturer publications are linked rather than redistributed. Component presence does not confirm availability or compatibility.

Document rendering, geometric consistency, source transcription and sample calculations are reviewed for this issue. No physical vehicle test or professional engineering approval is claimed. The release record states whether live browser interaction was observed; successful site deployment alone is not a WebGL interaction test.

09 | Source register

Current primary publications and retained project calculations are distinguished below. Each record identifies evidence and limits. Stock, price, lead time and project-specific approval require direct confirmation; none is inferred from a catalogue page.

C3DS01 | Atlas Tube ASTM A500 rectangular HSS section properties

Accessed 2026-10-18

Manufacturer-hosted PDF rows rechecked for five selected study sections: pages2,3,5,6. Nominal lb/ft, design wall, area, strong-axis I and S recorded. Table image for seat rail visually checked.

Limit: Table publication path2018; current access is not new certification, stock or rail application approval. Design properties and nominal weight serve different purposes.

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

C3DS02 | Steel Tube Institute HSS combined shear, axial and in-plane moment connection example

Accessed 2026-10-18

Technical example identifies combined member forces, local HSS joint checks and force distribution; no project detail copied.

Limit: Building-oriented example does not determine vehicle fatigue or adopted load criteria; member adequacy alone cannot release joints.

https://steeltubeinstitute.org/resources/hss-to-hss-connection-with-shear-axial-and-in-plane-moment-example/

C3DS03 | Plascore Base Panels

Accessed 2026-10-18

0.75in AA3.6-80 aluminum honeycomb panel construction and typical mass remain manufacturer published. Panel seams are coordinated geometrically here.

Limit: No panel span approved by this module. Phase3D body module separately reconciles typical rigidity sources; concentrated/rolling loads, shear, inserts and joints require OEM/engineer evidence.

https://plascore.com/composite-panels/base-panels/

C3DS04 | Retained Phase3C structure and Phase3D design basis

Accessed 2026-10-18

0%grade,36in gauge evaluation,13200kg regular upper study case,7040kg tare allocation,400in frame,216in bogie spacing and28in floor retained. D0 reproduces166member1913.259kg benchmark.

Limit: Project assumptions and prior calculations; not measured mass, supplier approval or load standard.

C3DG01 | Western Train FAQ

Accessed 2026-10-18

36-inch product offering, custom drive-truck regauging, custom parts and coach air brakes.

Limit: No published project-compatible unpowered bogie model, support height, load rating or offer.

https://www.westerntrain.com/faq

C3DG02 | Severn Lamb Jupiter

Accessed 2026-10-18

Jupiter identified: 914 mm gauge, 50 m minimum radius, 16 km/h maximum speed, enclosed-coach option.

Limit: Whole-train product claims; not ratings for this 65,100 kg train or standalone bogies. 914 versus 914.4 mm must be resolved.

https://www.severn-lamb.com/people-movers/tracked-trains/rail-trains/jupiter-2

C3DG03 | Railteco corporate passenger bogie catalog

Accessed 2026-10-18

RTKZ1 is a 762 mm forest-railway passenger bogie with 596 mm wheels; centre plate and sprung sidebearers share support; radial link mechanism and tread brake. Corporate site links Railway Spare Parts sales site.

Limit: Chinese catalog translated for technical summary. Does not establish 914.4 mm conversion, US application or support height.

https://cn.railteco.com/cplist.aspx?cptp=8&i=14

C3DG04 | Railteco RTKZ1 manufacturer sales technical page

Accessed 2026-10-18

Published RTKZ1: 3.75 t maximum axle, bogie tare <=1.5 t, 1,165 mm wheelbase, 596 mm wheel diameter, 40 km/h; brake lever ratio 4.84.

Limit: Page has erroneous 762m and RTKZ32 labels; product body/corporate page identify RTKZ1/762mm. Obtain controlled drawing; marketing certification labels not project approvals. Advertised four-month delivery is not current quotation/availability.

https://www.railwayspare-parts.com/sale-13771970-aar-standard-762m-gauge-passenger-train-bogie.html

C3DG05 | CFD two-axle passenger bogie

Accessed 2026-10-18

Manufacturer publishes 2.4 m wheelbase, 5-8 t bogie mass, 16 t axle maximum and 1000/(1067)/1435 mm gauges.

Limit: Useful scale comparison, rejected baseline; two bogies alone exceed coach tare allocation. No 914.4 mm offering established.

https://www.cfd.group/rolling-stock/railcar-bogie

C3DG06 | UTCRAS capabilities

Accessed 2026-10-18

Wheelset design/engineering, custom axles, truck assembly, NDT, spin and load testing.

Limit: Capability evidence only; small-gauge new-bogie design responsibility and commercial fit must be confirmed.

https://www.utcras.com/capabilities-1

C3DG07 | GHH-BONATRANS wheelsets

Accessed 2026-10-18

Tailored wheelsets taking bogie, brakes, drive and track into account; manufacturer describes all-gauge scope.

Limit: No selected small-diameter 914.4 mm part number, load envelope, MOQ or price.

https://www.ghh-radsatz.com/en/products-and-services/wheelset/

C3DG08 | Voith railway couplers

Accessed 2026-10-18

Type 430/530: compact/folding LRV family, compression/tension strength up to300kN. Type330: 800kN compression/600kN tension.

Limit: Family strengths are not working/fatigue ratings or dimensions for an offered assembly. Coupler length, height, strokes, articulation and mounting remain absent.

https://www.voith.com/corp-en/connection-components-couplings/railway-couplers.html

C3DG09 | Dellner semi-permanent people-mover coupler

Accessed 2026-10-18

People-mover semi-permanent coupler includes double-acting damper, pivot anchor and cable support in modular package.

Limit: No public dimensional or load rating used. Exact assembly and responsibility must be confirmed.

https://www.dellner.com/products/semi-permanent-couplers/semi-permanent-coupler-people-mover

C3DG10 | Atlantic Track rail section data sheet

Accessed 2026-10-18

40 ASCE: height/base3.5in, head1.875in, web25/64in; 60 ASCE: height/base4.25in, head2.375in, web31/64in.

Limit: Section identities only; no wheel compatibility, rail grade or track load acceptance. Ambiguous OCR head-depth entries not used.

https://atlantictrack.com/wp-content/uploads/2020/08/Rail-Section-Data-Sheet-Railroad-and-Transit.pdf

C3DB01 | Plascore - Base Panels

Accessed 2026-10-18

AA3.6-80 typical area masses .79/.84/.94/1.01 lb/ft² for .25/.5/.75/1 in. Aluminum skins .020 in, core nominal 3.6 lb/ft³. Typical rigidity 46,858 and 71,654 lb·in² per inch width for .75 and 1 in; four-point quarter-point test, 20 in span and 3 in specimen width.

Limit: SI table heading appears malformed. No verified core orientation, separated shear rigidity, minimum values, floor span or rolling-load rating. Published rigidity differs from marine brochure.

https://plascore.com/composite-panels/base-panels/

C3DB02 | Plascore - Engineered Honeycomb marine brochure, p5

Accessed 2026-10-18

Screenshot confirms rigidity unit lb·in² / inch width; AA3.6-80 .75 in =44,000 and 1 in =71,000; nominal weights .94 and1.01 lb/ft². Twenty-inch span and three-inch width test footnote.

Limit: Undated public brochure uses different rigidity values from current webpage. Lower values used only for comparison; supplier must reconcile exact grade, direction, lot minimum and shear properties.

https://plascore.com/download/1951/

C3DB03 | Gerflor - Tarabus Sirius technical data

Accessed 2026-10-18

NT thickness2.25±.15mm, mass2.20±.15kg/m²; R10 slip classification; FMVSS302 and ASTM E648 Class1 claims.

Limit: No installed composite-floor approval or farm-mud slip result; exact adhesive and complete fire basis remain held.

https://cdn.gerflor.com/media/2/38529/tarabus%20sirius%20-%20technical%20datasheet.pdf

C3DB04 | Plascore - Value Add

Accessed 2026-10-18

Manufacturer offers edge framing and potted/through-fastened insert approaches.

Limit: No project insert, edge-closure or anchor rating. No bolted or bonded detail released.

https://plascore.com/composite-panels/value-add/

C3DB05 | Pilkington ATS-129 - Properties of Soda-Lime Silica Float Glass, April2025

Accessed 2026-10-18

Glass density2500kg/m³.

Limit: Density supports mass only. Safety treatment, laminate/IGU construction, thickness, emergency-egress retention and rail suitability unselected.

https://www.pilkington.com/-/media/pilkington/site-content/usa/window-manufacturers/technical-bulletins/ats-129---properties-of-soda-lime-silica-float-glass.pdf

C3DB06 | Armacell - ArmaFlex Rail SD technical specifications

Accessed 2026-10-18

Conductivity expression[40+.1T+.0009T²]/1000W/mK; E162/E662 claim limited3-13mm; required product adhesives and UV protection stated.

Limit: No density on reviewed page; no extension of fire claim to thicker stacks or entire coach. Thermal bridges and US application unverified.

https://www.armacell.com/en-NL/product/technical-specs/armaflex-rail-sd

C3DB07 | Konvekta Rail HVAC overview08/2021, HVAC7403 p33 and5401 p23

Accessed 2026-10-18

HVAC7403 mass520kg, envelope3100×1600×415mm without fasteners. HVAC5401 mass275kg (optional aluminum compressor240kg), envelope1500×1500×370mm. Both400VAC;24VDC controls.

Limit: Old public catalog rechecked; no current stock, application approval, US power/ambient performance or mount loads. Mass reference is not selected equipment.

https://www.konvekta.de/fileadmin/user_upload/docs/datenblaetter/schiene/Produktuebersicht_Schiene_dt-engl_eBook_WEB_0821.pdf

C3DB08 | Sika Sikaflex252 US datasheet04.01,April2025

Accessed 2026-10-18

Vehicle assembly adhesive; uncured density1.20kg/L; actual-substrate/condition tests required; typical lap-shear2.5MPa.

Limit: Typical strength is not allowable stress; installed volume, coatings, peel/creep/fatigue/fire evidence unknown. Not selected floor-finish adhesive.

https://industry.sika.com/dam/dms/us01/o/sikaflex-252.pdf

C3DB09 | ASTM C393/C393M-20 public scope

Accessed 2026-10-18

Published scope distinguishes core shear/beam stiffness, facing strength and separated flexural/shear stiffness test methods.

Limit: Public scope only; full standard not obtained. Laboratory and engineer specify current edition, specimens, conditioning and acceptance; no assumed legal applicability.

https://store.astm.org/c0393_c0393m-20.html

C3DB10 | ASTM D7250/D7250M-20 public scope

Accessed 2026-10-18

Practice identifies calculation of sandwich beam flexural and shear stiffness from test results.

Limit: Public scope only; no full standard or test procedure reproduced. No strength allowable established.

https://store.astm.org/d7250_d7250m-20.html

C3DB11 | BraunAbility NCL1000 Century application guide,3June2026

Accessed 2026-10-18

NCL1000IB3754HB-2:385lb mass,1000lb rating,minimum45.85in doorway,50.96in overallwidth,18in threshold depth.

Limit: Reference dimensions only; motor-vehicle mounting kits are not rail mounting approval. Actual mount CAD/reactions, stow clearance and independent rescue still held.

https://www.braunability.com/content/dam/braunability/docs/comm/na/app-guide/updated-dpa-lift-app-guides/Lifts-Comm-Century-1000-Guide-6-3-26.pdf

C3DP01 | HydroAct hydraulic brake calipers

Accessed 2026-10-18

HSC1P10N spring floating caliper:20kN manufacturer brake force per disc, two hydraulic release circuits, disc width up to45mm,approx15kg.

Limit: Force convention, release pressure, force curves, repeated-service duty, disc and mounting configuration absent.

https://rail.knorr-bremse.com/media/2000_products/2900d_product_broschures/p-1236_hydroact.pdf

C3DP02 | Electro-hydraulic module EHM

Accessed 2026-10-18

Two bogies independently activated;24VDC power,48VDC train-control wire;1100x510x500mm;approx90kg;160bar permissible system pressure.

Limit: Not a specified operating/release pressure, pump flow or current; project configuration unresolved.

https://www.hanning-kahl.com/products/hydraulic-brake-systems/electro-hydraulic-module.html

C3DP03 | HCU hydraulic unit

Accessed 2026-10-18

Configurable modular manifold, accessible adjustable/sealable relief valves, filtration and current-reduction functions.

Limit: No public project flow, selected pressure, voltage/current or emergency valve circuit.

https://www.hanning-kahl.com/products/brakemaster/hydraulic-unit-of-the-hcu-series.html

C3DP04 | ECU control unit

Accessed 2026-10-18

Nominal24-110VDC; hardwired and CAN/MVB/Ethernet interfaces; manufacturer states replacement of HEYxPxG/HEY-C from2028.

Limit: No configured current, accepted timeout or certificate/application approval inferred.

https://www.hanning-kahl.com/products/brakemaster/control-unit-of-the-ecu-series.html

C3DP05 | Auxiliary release devices

Accessed 2026-10-18

Auxiliary devices release spring actuators via a second hydraulic circuit; hand/powered variants described.

Limit: Second release circuit is not a second independent braking application path.

https://www.hanning-kahl.com/products/hydraulic-brake-systems/auxiliary-release-devices.html

C3DP06 | Rail HVAC product overview,7403 and5401

Accessed 2026-10-18

7403:30kW cooling at35C outdoor/30C indoor,30kW heater,400VAC/24VDC control,520kg;5401:16kW at40C/40C,7.2kW heater,400VAC/24VDC,275kg.

Limit: Catalog2021 still publicly available; current production variant,60Hz,actual43.3C/23.9C capacity,input amps/MCA/MOCP/start not verified.

https://www.konvekta.de/fileadmin/user_upload/docs/datenblaetter/schiene/Produktuebersicht_Schiene_dt-engl_eBook_WEB_0821.pdf

C3DP07 | NW-64-CN passenger-car condenser

Accessed 2026-10-18

Nominal5-ton condenser family; usual480V three phase;240V variants; configurations differ by compressor.

Limit: No whole-coach duty, current, start or matched mass verified. Filename/model family not an orderable matched package.

https://www.nwrail.com/nwre_products/64cn.html

C3DP08 | NW-25100 electrical locker

Accessed 2026-10-18

Typical480V three-phase system;120/208V distribution; staged HVAC;24V controls; typical emergency-light relay circuit20A at12V or13A at28V; custom options.

Limit: This is passenger-utility distribution, not a railway safety brake controller. Current values are circuit ratings, not actual loads or battery duration.

https://www.nwrail.com/nwre_products/25100_index.html

C3DP09 | Mark6 generator specifications S32.25

Accessed 2026-10-18

6-42T4F/6-80T4:38/70kW continuous;60Hz standard240/480V three phase. Alternator motor-start capability124/270skVA at35% voltage dip forT240/T480. Reference80F/500ft/50%RH.

Limit: Site derating, installed configuration and acceptable load voltage dip must be returned. Alternator starting figure not complete engine/genset transient guarantee.

https://stadcogen.com/wp-content/uploads/2025/05/S32.25-Mark-6-Specs.pdf

C3DP10 | HEP trainline configurations

Accessed 2026-10-18

Multiple480V trainline arrangements described, requiring source isolation or deliberate synchronization for multiple generators; system protection important.

Limit: Full-sized trainline conductor/connector examples are not this small-train cable design; generic tourist kW examples not our loads.

https://www.nwrail.com/HEP_config.html